



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

January 21, 2026

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Explain the bias–variance trade-off in supervised learning. Define bias and variance, describe their relation to model complexity, and explain how they lead to underfitting and overfitting.

Exercise 2 (7 marks)

Compare Logistic Regression and Support Vector Machines. Discuss their loss functions, output interpretation, margin, regularization, and mention one scenario where each method is preferable.

Exercise 3 (2 marks)

Consider the following Python code snippet:

```
1 X = dataset[['feature1', 'feature2']].values
2 y = dataset['label'].values
3
4 scaler = StandardScaler()
5 X_scaled = scaler.fit_transform(X)
6
7 model = LogisticRegression(C=0.01)
8 model.fit(X_scaled, y)
9
10 predictions = model.predict(X_scaled)
11 accuracy = (predictions == y).mean()
```

1. Describe briefly what the code does and what learning task it is solving.
2. Explain the effect of the parameter $C=0.01$ in this context.
3. Is computing the accuracy on `X_scaled` appropriate to evaluate performance? Why or why not?

1. The code solves a classification problem using logistic regression. It standardizes two input features, trains a logistic regression model on the scaled data, predicts labels on the training set, and computes the training accuracy.
2. C is the inverse of the regularization strength. Setting $C=0.01$ applies strong regularization, shrinking the model weights, reducing variance, and increasing bias, which may lead to underfitting if too strong.
3. No. The accuracy is computed on the training data, so it does not provide a reliable estimate of the model's generalization performance. To properly evaluate the classifier, accuracy should be measured on a separate validation or test set (or via cross-validation). Additionally, the scaler should be fitted only on the training data to avoid data leakage.³

Exercise 4 (2 marks)

Tell whether the following statements about Kernel Methods are true or false. Briefly motivate your answers.

1. If the feature mapping $\phi(x)$ is explicitly known, there is no point in using a kernel (dual) representation.
2. The function $k(x, x') := \exp(-\|x - x'\|^2)$ is a valid kernel.
3. The function $k(x, x') := x^\top A x'$ with $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $x, x' \in \mathbb{R}^2$ is a valid kernel.
4. Knowing only the Gram matrix K of the training set is sufficient to compute predictions on a new test point x^* using a dual kernel method.

1. **False.** Even with an explicit feature map, the dual/kernel form can still be preferable (e.g., from computational perspective, depending on n vs. feature dimension).
2. **True.** This is the Gaussian/RBF kernel.
3. **False.** Although A is symmetric, it is not PSD (it has a negative eigenvalue), hence k is not a valid kernel.
4. **False.** To predict at x^* , one also needs the vector of kernel evaluations $k(x^*, x_i)$ (and the learned dual coefficients); K alone is insufficient.

Exercise 5 (2 marks)

Tell if the following statements about the VC dimension are true or false. Briefly motivate your answers.

1. Even if a hypothesis space H contains infinitely many hypotheses, its VC dimension is always finite.
2. The VC dimension of H is at least k if and only if H shatters *at least one* subset of the instance space of cardinality k .
3. The VC dimension of a logistic-regression classifier using features (x_1, x_2) is smaller than the VC dimension of a logistic-regression classifier using $(x_1, x_2, x_1^2, x_2^2, x_1x_2)$.
4. The VC dimension of a hypothesis space depends on the size of the training set.

1. **False.** There exist infinite hypothesis spaces with infinite VC dimension (they can shatter sets of any finite size).
2. **True.** $VC(H) \geq k$ iff there exists a shattered set of size k .
3. **True.** Logistic regression is linear in the chosen feature space; adding features enlarges the class and increases the VC dimension with the feature dimension.
4. **False.** VC dimension is a property of H only; it does not depend on training/test set size.

Exercise 6 (2 marks)

You are a data scientist for a *bike-sharing* service. You have historical logs containing, for each user and week, aggregated activity (e.g., #rides, total minutes, average distance, typical time-of-day, city, subscription tier) and service-quality indicators (e.g., average nearby-bike availability, reported issues, support contacts).

1. You want to predict whether a user will make **zero rides in the next 30 days**.
2. Every Monday you want to send each user **one** offer among K alternatives (e.g., discount, free minutes, temporary upgrade, no offer).

Model the two problems described above as ML problems, specifying the key elements such as features and target or states, actions, and reward, and mention suitable methods to solve them.

1. Classification (supervised learning).

- **Instances:** one example per (user, reference week).
- **Features:** recent-window aggregates and trends of usage (rides/minutes/distance), recency, time-of-day patterns, city, subscription tier, service-quality indicators (availability, issues, support contacts), calendar/seasonality if available.
- **Target:** binary label indicating whether the user makes zero rides in the next 30 days.
- **Methods:** logistic regression or support vector machines.

2. Multi-Armed Bandit (sequential decision making).

- **State / context (optional):** user profile + recent behavior/service-quality features available on Monday.
- **Actions:** choose one offer among the K alternatives for each user.
- **Reward:** observed after the week (e.g., 1 if at least one ride occurred, 0 otherwise; or rides/revenue).
- **Methods:** UCB or Thompson Sampling (eventually contextual).

Exercise 7 (4 marks)

Assume you need to learn the parameters of a binary perceptron classifier with bias. Start from $\mathbf{w}^{(0)} = (1 \ 1 \ 0)^\top$ and use learning rate $\alpha = 1$.

You are given the following dataset (in the reported order):

$$\begin{array}{ll} \mathbf{x}_1 = (2 \ 2)^\top & t_1 = 1 \\ \mathbf{x}_2 = (2 \ -2)^\top & t_2 = -1 \\ \mathbf{x}_3 = (-2 \ 2)^\top & t_3 = -1 \\ \mathbf{x}_4 = (-2 \ -2)^\top & t_4 = 1 \end{array}$$

1. Compute the vector $\mathbf{w}$ after running one full pass of the online perceptron algorithm on the dataset.
2. With the obtained $\mathbf{w}$, which training points are misclassified?
3. Can you tell if the perceptron training procedure would converge eventually on this dataset? Motivate your answer.

1. Online update: if $\hat{t}_n \neq t_n$, then $\mathbf{w} \leftarrow \mathbf{w} + t_n \tilde{\mathbf{x}}_n$ (where $\tilde{\mathbf{x}}_n$ is the augmented input vector with bias term).

$$n = 1 : \tilde{\mathbf{x}}_1 = (1, 2, 2), a_1 = 3 \Rightarrow \hat{t}_1 = 1 = t_1 \Rightarrow \mathbf{w} = (1, 1, 0)$$

$$n = 2 : \tilde{\mathbf{x}}_2 = (1, 2, -2), a_2 = 3 \Rightarrow \hat{t}_2 = 1 \neq t_2 \Rightarrow \mathbf{w} = (1, 1, 0) - (1, 2, -2) = (0, -1, 2)$$

$$n = 3 : \tilde{\mathbf{x}}_3 = (1, -2, 2), a_3 = 6 \Rightarrow \hat{t}_3 = 1 \neq t_3 \Rightarrow \mathbf{w} = (0, -1, 2) - (1, -2, 2) = (-1, 1, 0)$$

$$n = 4 : \tilde{\mathbf{x}}_4 = (1, -2, -2), a_4 = -3 \Rightarrow \hat{t}_4 = -1 \neq t_4 \Rightarrow \mathbf{w} = (-1, 1, 0) + (1, -2, -2) = (0, -1, -2)$$

$$\boxed{\mathbf{w} = (0 \ -1 \ -2)^\top.}$$

2. Check with $\mathbf{w} = (0, -1, -2)$:

$$\mathbf{x}_1 : \mathbf{w}^\top (1, 2, 2) = -6 \Rightarrow \hat{t}_1 = -1 \neq t_1$$

$$\mathbf{x}_2 : \mathbf{w}^\top (1, 2, -2) = 2 \Rightarrow \hat{t}_2 = 1 \neq t_2$$

$$\mathbf{x}_3 : \mathbf{w}^\top (1, -2, 2) = -2 \Rightarrow \hat{t}_3 = -1 = t_3$$

$$\mathbf{x}_4 : \mathbf{w}^\top (1, -2, -2) = 6 \Rightarrow \hat{t}_4 = 1 = t_4$$

Misclassified: $\boxed{\mathbf{x}_1, \mathbf{x}_2}$.

3. The labels form an XOR pattern: positives are $(2, 2)$ and $(-2, -2)$, negatives are $(2, -2)$ and $(-2, 2)$. The convex hulls of the two classes intersect (both contain the origin), hence the data are not linearly separable. Therefore the perceptron algorithm will not converge.

Exercise 8 (4 marks)

Consider the following episode obtained by an agent interacting with an MDP having three states $\mathcal{S} = \{A, B, C\}$ and two actions $\mathcal{A} = \{l, r\}$,

$$(A, l, 0) \rightarrow (B, r, 4) \rightarrow (C, l, -2) \rightarrow (B, l, -1) \rightarrow (A, r, 2) \rightarrow (B, r, 0) \rightarrow (A, l, 1) \rightarrow (C, l).$$

Consider an initial state-action values $Q(s, a) = 0$ for every state-action pair, a learning rate $\alpha = 0.5$, and a discount factor $\gamma = 1$. Answer to the following questions providing adequate motivations (break ties using the first action listed in the action set).

1. Execute the *Q-learning* algorithm on the given episode.
2. Execute the *SARSA* algorithm on the given episode.
3. Provide the best deterministic policy according to the output of *SARSA*. Is it different from the one provided by *Q-learning*?

1. The formula for Q-learning is $Q(s, a) \leftarrow Q(s, a) + \alpha(R + \gamma \max_{a'} Q(s', a') - Q(s, a))$, which in our case is:

$$Q(A, l) = 0 + 0.5(0 + 1 \cdot 0 - 0) = 0$$

$$Q(B, r) = 0 + 0.5(4 + 1 \cdot 0 - 0) = 2$$

$$Q(C, l) = 0 + 0.5(-2 + 1 \cdot \max\{Q(B, l), Q(B, r)\} - 0) = 0.5(-2 + 2) = 0$$

$$Q(B, l) = 0 + 0.5(-1 + 1 \cdot \max\{Q(A, l), Q(A, r)\} - 0) = -1/2$$

$$Q(A, r) = 0 + 0.5(2 + 1 \cdot \max\{Q(B, l), Q(B, r)\} - 0) = 0.5(2 + 2) = 2$$

$$Q(B, r) = 2 + 0.5(0 + 1 \cdot \max\{Q(A, l), Q(A, r)\} - 2) = 2$$

$$Q(A, l) = 0 + 0.5(1 + 1 \cdot \max\{Q(C, l), Q(C, r)\} - 0) = 1/2$$

2. The formula for SARSA is $Q(s, a) \leftarrow Q(s, a) + \alpha(R + \gamma Q(s', a') - Q(s, a))$, where a' is the action observed in the episode. In our case:

$$Q(A, l) = 0 + 0.5(0 + 1 \cdot Q(B, r) - 0) = 0$$

$$Q(B, r) = 0 + 0.5(4 + 1 \cdot Q(C, l) - 0) = 2$$

$$Q(C, l) = 0 + 0.5(-2 + 1 \cdot Q(B, l) - 0) = -1$$

$$Q(B, l) = 0 + 0.5(-1 + 1 \cdot Q(A, r) - 0) = -1/2$$

$$Q(A, r) = 0 + 0.5(2 + 1 \cdot Q(B, r) - 0) = 2$$

$$Q(B, r) = 2 + 0.5(0 + 1 \cdot Q(A, l) - 2) = 1$$

$$Q(A, l) = 0 + 0.5(1 + 1 \cdot Q(C, l) - 0) = 0$$

3. The policy prescribed by SARSA is to take the greedy action in each state (ties broken with the first action):

$$\pi_{\text{SARSA}}(A) = r, \quad \pi_{\text{SARSA}}(B) = r, \quad \pi_{\text{SARSA}}(C) = r$$

since $Q_{\text{SARSA}}(C, r) = 0 > Q_{\text{SARSA}}(C, l) = -1$. For Q-learning we get $\pi_{\text{QL}}(A) = r$, $\pi_{\text{QL}}(B) = r$, while in C we have $Q_{\text{QL}}(C, l) = Q_{\text{QL}}(C, r) = 0$, hence $\pi_{\text{QL}}(C) = l$ by tie-breaking. Therefore, the policies are different (in state C).